

Seema Sultana

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EDUCATION

- **HKBK College Of Engineering - VTU affiliated** Bengaluru, India
Bachelor of Engineering - Computer Science And Engineering; GPA: 8.25 sept 2023 - present
- **Presidency PU College** Bengaluru, India
Class 12th - State Board; Grade: 92.5 2021
- **SVN English High School** Bengaluru, India
Class 10th - State Board; Grade: 91.04 2019

SKILLS SUMMARY

- **Languages:** Python, JAVA, SQL, Typescript, Javascript
- **Frontend:** React.js, Next.js, Tailwind CSS
- **Backend:** FastAPI, Django, Flask, REST APIs
- **Database:** PostgreSQL, MySQL, MongoDB
- **Frameworks:** NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, XGBoost, LangChain
- **Platforms:** Linux, Web, Windows, Arduino, AWS, IBM Cloud
- **Tools:** AWS (EC2, S3, Lambda, IAM, Bedrock, SageMaker), Git, GitHub, Docker
- **Soft Skills:** Leadership, Event Management, Writing, Public Speaking, Time Management

PROJECTS

- **Privacy-Preserving Federated Learning for IoT Intrusion Detection :** Project Link
Built a PyTorch/Flower federated learning pipeline on UNSW-NB15 with Differential Privacy, Secure Aggregation, and adaptive communication, evaluating accuracy, privacy, convergence, and communication overhead.
Tech: Python, PyTorch, Flower, Scikit-learn, Opacus (Differential Privacy), Secure Aggregation, Adaptive Communication.
- **FactSphere — Hallucination-Aware Multi-Agent Search Intelligence :** Project Link
Developed a 4-agent AI fact-checking system integrating multi-search retrieval, evidence deduplication, Groq LLM trust scoring, and a Chrome Extension, achieving 91% fabricated-claim detection accuracy.
Tech: Python, Flask, Groq API, Llama 3.3 70B, BeautifulSoup, SentenceTransformers, ChromaDB, scikit-learn, Chrome Extension API (2026)
- **Customer Churn Prediction using Explainable Machine Learning :** Project Link
Built an explainable AI pipeline using SHAP to identify key churn drivers and generated actionable business insights through an interactive Streamlit dashboard, enabling data-driven customer retention strategies.
Tech: Python, Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Matplotlib, Seaborn, Streamlit

EXPERIENCE

- **AWS Student builder group** On Campus
Co- lead (Full-time) May 2026 - present
 - **Study Jam:** Completed multiple AWS certifications to strengthen my understanding of technologies
 - **Projects:** Worked on Cognitive Load Aware Multi Agent AI system using Multiple AWS tools Project Link
- **Embrizon Tecnologies - Data Science Internship** Remote
Data Science with AI - Certificate link Feb - Apr 2026
 - **Project Course - Customer Churn Prediction system:** Performed data cleaning, feature engineering, and trained classification models including Logistic Regression, Random Forest, and XGBoost.

PUBLICATIONS

Analysis and Implementation of Privacy-Preserving Federated Learning Frameworks for IoT-Based Distributed Data Analytics : Work in Progress Paper - IEEE Publishing in late 2026.
Tech : Python, PyTorch, Flower (Federated Learning), Scikit-learn, Pandas, NumPy, Opacus (Differential Privacy), Flower Secure Aggregation, Adaptive Communication.

HONORS AND AWARDS

- 1st Place – ELECTROLYTHON National Level Hackathon - Sept, 2025
- Top 6 Finalist – Gen-AI Hackathon - May, 2026
- Shortlisted – Google Build with AI (Bangalore) - June, 2026

VOLUNTEER EXPERIENCE

- **AWS Student Builder Group - Vice President/Operations Lead** Bengaluru, India
Organized workshops and technical events through AWS, engaging students and developers July 2026 - Present
- **IEEE Computer Society Bangalore Chapter (IEEE CSBC)** Bengaluru, India
Managing the official IEEE CSBC website Coordinated IEEE CS PRO 2026, Girl Geek. Jan 2026 - Present